

REAL-TIME MINE SAFETY MONITORING AND GAS DETECTION SYSTEM

Major Diploma Project Documentation

Electronics & Communication Engineering



Actual assembled project hardware prototype

Vattam Madhu Kumar

AANM & VVRSR Polytechnic College, Gudlavalleru

GitHub: github.com/MadhuKumar-003

Contents

1. Introduction
2. Problem Statement
3. Project Objectives
4. Functional System Architecture
5. Project Hardware Prototype
6. Gas Sensing Section
7. Arduino / ESP32 Controller
8. 16x2 LCD Display
9. MQ-135 Gas Sensor
10. MQ-7 Carbon Monoxide Sensor
11. Wireless Communication
12. Alarm and Warning System
13. SD-Card Data Logging
14. Power Supply and Backup
15. Arduino IDE and Software Development
16. Working Principle
17. Testing and Validation
18. Project Responsibilities
19. Engineering Skills Demonstrated
20. Limitations and Documentation Boundaries
21. Future Improvements
22. Repository Structure
23. Supporting Technical References
24. Conclusion
25. Author
26. Project Image
27. Functional System Diagram

1. Introduction

Safety monitoring is an important requirement in environments where hazardous gases may accumulate. A monitoring unit can provide early warning by sensing gas conditions and presenting the information to an operator while also supporting remote alerts and data logging.

The project was developed to demonstrate how sensors, embedded controllers, displays, alarms, wireless communication modules, and power-management elements can be integrated into a working prototype.

The physical prototype is documented through project photographs, while the supporting component notes provide background on the key hardware and software used in development.

2. Problem Statement

Mining-oriented environments can present hazardous conditions due to the presence of combustible or toxic gases. Relying only on manual observation can delay recognition of an unsafe condition.

- monitor gas-related sensor inputs
- process sensor information using an embedded controller
- provide local status information
- activate warning indicators
- support remote communication
- record monitoring information
- consider continuity and reliability through backup power and safety concepts

3. Project Objectives

1. Develop a real-time gas monitoring prototype for a mine-safety application.
2. Monitor CH₄, CO, and H₂S using gas-sensing hardware.
3. Process sensor information using Arduino/ESP32-based embedded control.
4. Provide local information through a display.
5. Generate audible and visual warning indications.
6. Support remote alerting using GSM/LoRa as described in the project record.
7. Provide SD-card data logging.
8. Include battery-backup considerations for improved continuity.
9. Include self-test and fail-safe alarm concepts.
10. Gain practical experience in sensor interfacing, embedded hardware integration, testing, troubleshooting, and team leadership.

4. Functional System Architecture

The system can be viewed as six functional layers: gas sensing, embedded processing, user indication, local alarm, communication, and logging/power support.

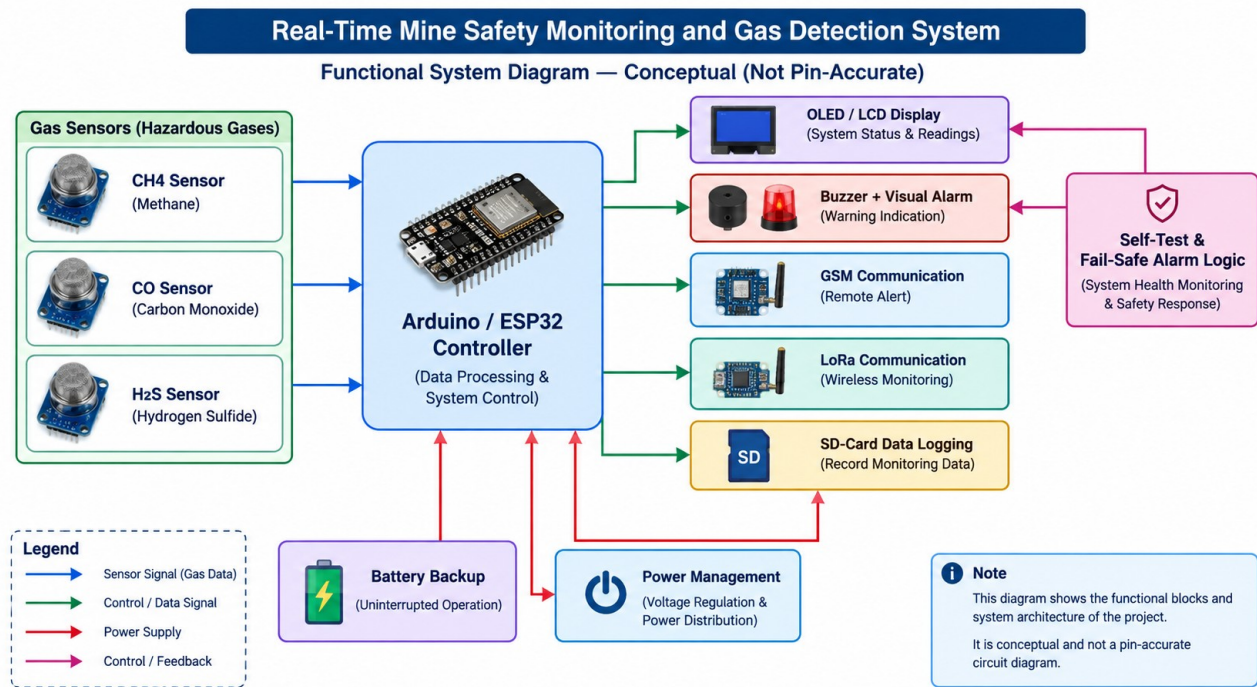


Figure 1. Functional system architecture - conceptual, not pin-accurate.

The graphical diagram is intentionally functional rather than a fabrication-ready schematic. Exact pin assignments are not asserted because the supplied materials do not provide the final pin-level circuit.

5. Project Hardware Prototype

The photograph below shows the actual assembled prototype used for the major diploma project. The two mounting boards present the integrated sensors, controller boards, display modules, power circuitry, communication/interface modules, and supporting wiring.



Figure 2. Actual project hardware prototype.

This photograph should remain prominent in the GitHub repository because it provides direct visual evidence that the project was physically assembled and integrated.

6. Gas Sensing Section

6.1 Methane (CH₄)

The project record identifies methane as one of the monitored hazardous gases. The gas-sensing section provides environmental input to the embedded controller, which uses sensor information as part of the monitoring and warning logic.

6.2 Carbon Monoxide (CO)

The project record identifies CO as a monitored gas. The supplied MQ-7 technical note describes MQ-7 as a carbon-monoxide sensor whose electrical resistance changes with CO concentration. The note also describes analog and digital outputs that can be read by an Arduino for monitoring and threshold-based alerting.

6.3 Hydrogen Sulfide (H₂S)

H₂S is included in the original project record as a monitored hazardous gas. The current supporting component notes do not provide an H₂S-specific sensor document, so this report does not assign a specific sensor model or pin mapping to H₂S.

7. Arduino / ESP32 Controller

The embedded controller is the central processing block of the system. It receives sensor information, performs monitoring logic, controls display and alarm outputs, and coordinates communication and logging functions.

The supplied Arduino Uno note documents the board as an ATmega328P-based microcontroller with 14 digital I/O pins, 6 analog inputs, 6 PWM-capable digital pins, 5 V operating logic, 16 MHz clock operation, and UART, SPI, and I2C capabilities.

- A0-A5 are documented as analog inputs suitable for reading sensor voltages.
- D0-D13 are documented as digital input/output pins.
- D0 and D1 support serial TX/RX functions.
- The board can be programmed through the Arduino IDE.

Exact pin assignments for the final project are intentionally not specified here because they are not available in the supplied project documents.

8. 16x2 LCD Display

The supplied LCD note documents a 16x2 character LCD, meaning 16 columns and 2 rows, with the ability to display up to 32 characters at a time. It describes a standard 16-pin interface and Arduino interfacing through the LiquidCrystal library.

Pin	Function	Purpose
1	GND	Ground reference
2	VCC	Supply
3	V0/VEE	Contrast control
4	RS	Register Select
5	R/W	Read/Write control
6	E	Enable
7-14	Data	4-bit or 8-bit data
15	LED+	Backlight supply
16	LED-	Backlight ground

For this project, the display serves as the local user interface for presenting monitoring and system-status information.

9. MQ-135 Gas Sensor

The supplied MQ-135 note describes the MQ-135 as a gas-sensing module used to monitor the presence of multiple gases. It explains that the sensing element is chemiresistive and that changes in sensor resistance are converted into a voltage that can be read by a microcontroller.

- analog output behavior
- digital threshold output
- adjustable sensitivity
- need for sensor warm-up
- need for calibration
- Arduino interfacing through analog or digital input

For an engineering implementation, MQ-series readings should be treated as sensor outputs that require appropriate calibration before being interpreted as quantitative gas concentrations.

10. MQ-7 Carbon Monoxide Sensor

The supplied MQ-7 note specifically describes MQ-7 as a carbon monoxide sensor. It explains that CO changes the sensing element electrical resistance and that the resulting change can be measured by a microcontroller.

- VCC - supply
- GND - ground
- AO - analog output
- DO - digital threshold output

The supplied Arduino interfacing description uses `analogRead()` for analog measurement and `digitalRead()` for threshold-based detection. The note also emphasizes calibration and conditioning for reliable measurements.

11. Wireless Communication

11.1 GSM / LoRa - Project Record

The original project record describes GSM and LoRa integration for alerts. These communication paths were included as part of the project remote-alert concept.

11.2 Wi-Fi - Supporting Technical Reference

A separate Wi-Fi technical note was provided with the project materials. It explains that Wi-Fi modules provide wireless data communication and commonly interface with microcontrollers through UART, SPI, or other digital interfaces. The note also describes client and access-point operation and remote monitoring use cases.

Because the original project record explicitly lists GSM/LoRa while the supplied component notes describe Wi-Fi, the final documentation keeps these technologies separated rather than claiming that all three were used simultaneously in the final hardware.

12. Alarm and Warning System

The project record describes buzzer and strobe/visual warning outputs. These indicators form the local safety-notification layer. The controller can use the sensor-monitoring logic to produce a warning state, which can then drive an audible buzzer and a visual or strobe indicator.

13. SD-Card Data Logging

The project record states that SD-card logging was integrated. Data logging provides a way to retain monitoring information for later review, system testing, and record keeping. The supplied documents do not provide the exact SD-card wiring or file format, so no pin mapping or software schema is claimed here.

14. Power Supply and Backup

The supplied 12V 1A SMPS note describes a switched-mode power supply with a regulated 12 V DC output at up to 1 A. It documents 100-265 VAC input, 12 V DC output, and over-voltage/over-current protection features.

The project record also states that battery backup was added to improve reliability and continuity. The exact final conversion stages and power rails of the completed prototype are not established by the supplied files, so this document treats the SMPS as a documented power-supply reference and the battery as a project-level backup-power concept.

15. Arduino IDE and Software Development

The supplied Arduino IDE note describes the IDE as the software environment used to write, verify/compile, and upload sketches to Arduino hardware.

- Sketches use .ino source files.
- `setup()` is used for initialization and one-time configuration.
- `loop()` runs repeatedly and contains the main operational logic.
- Board and port selection are part of the upload workflow.
- The Serial Monitor supports debugging and monitoring.
- Libraries can be added to extend hardware functionality.

16. Working Principle

The system follows this general operating sequence:

1. Gas sensors monitor the surrounding environment.
2. Sensor outputs are received by the embedded controller.
3. The controller processes the sensor information.
4. Monitoring information is presented on the display.
5. When the programmed safety condition is met, alarm outputs are activated.
6. Remote communication can be used for alert transmission.
7. Monitoring data can be recorded to SD card.
8. Battery-backup and self-test concepts improve continuity and reliability.

17. Testing and Validation

Testing was approached at module-integration and system-prototype levels. The project materials support testing of sensor interfacing, controller operation, display operation, alarm outputs, communication-module integration, SD-card logging, battery-backup behavior, self-test concepts, and overall hardware integration.

The supplied component notes also highlight the importance of calibration and warm-up for MQ-series gas sensors. This is especially important when sensor readings are used for safety-related threshold decisions.

No numerical test results are stated in this document because they were not provided in the supplied project materials.

18. Project Responsibilities

The original project record states that the project team was led by Madhu Kumar. The documented project responsibilities include:

- project-team leadership

- real-time gas monitoring development
- Arduino/ESP32 integration
- display integration
- alarm implementation
- GSM/LoRa integration
- SD-card logging
- battery-backup implementation
- self-test and fail-safe concepts
- overall system integration

19. Engineering Skills Demonstrated

- Embedded hardware integration
- Sensor interfacing
- Analog/digital signal reading
- Microcontroller-based control
- Display interfacing
- Wireless communication concepts
- Data logging
- Power-system awareness
- Hardware testing
- Troubleshooting
- Project leadership

20. Limitations and Documentation Boundaries

This report intentionally avoids claiming details that were not present in the supplied files.

- The exact final pin-to-pin circuit is not available.
- The exact sensor model used for every gas in the completed prototype is not established by the supplied notes.
- Exact threshold values and calibration coefficients are not provided.
- Exact GSM/LoRa module models and pin mappings are not provided.
- Exact SD-card storage format is not provided.
- The Wi-Fi note is a related technical reference and does not replace the project stated GSM/LoRa integration.

The included system architecture is therefore a functional/conceptual diagram, not a fabrication-ready schematic.

21. Future Improvements

- Dedicated PCB design instead of prototype wiring
- Improved sensor calibration

- More robust gas-sensor compensation
- Integrated power-management circuitry
- Enclosure and mechanical protection
- Real-time web dashboard
- Mobile alert interface
- Cloud-based data storage
- Additional environmental sensing
- Automated fault detection

22. Repository Structure

```
mine-safety-monitoring-system/  
|-- README.md  
|-- PROJECT_DOCUMENTATION.md  
|  
|-- images/  
|   |-- project-hardware-rotated.jpg  
|   |-- system-functional-diagram.png  
|  
|-- src/  
|   |-- [actual Arduino/ESP32 source code]  
|  
|-- circuit/  
|   |-- [verified circuit/schematic, if available]  
|  
|-- docs/  
    |-- Mine_Safety_Monitoring_Project_Documentation.pdf
```

Only upload actual source code and verified schematics. Do not add invented pin mappings or fabricated measurements.

23. Supporting Technical References

Reference document	Content used as technical background
LCD (Arduino)	16x2 LCD structure, pins, working principle, and Arduino interfacing.
MQ-135 (Arduino)	Gas-sensor operation, analog/digital outputs, calibration, and Arduino interfacing.
MQ-7 Sensor (Arduino)	Carbon-monoxide sensing, pin configuration, and Arduino interfacing.
Wi-Fi (Arduino)	Wireless module concepts, network operation, and Arduino/Raspberry Pi interfacing.
12V 1A SMPS Power Supply	Regulated power-supply characteristics, operation,

Real-Time Mine Safety Monitoring and Gas Detection System and protection features.

Arduino IDE

Sketch development, compile/upload workflow, setup/loop structure, libraries, and Serial Monitor.

Arduino Uno

ATmega328P platform, pin functions, power, memory, peripherals, and applications.

These documents can be kept under a GitHub docs/reference/ folder if desired.

24. Conclusion

The Real-Time Mine Safety Monitoring and Gas Detection System represents a major diploma project focused on practical electronics and embedded-system integration.

The project brings together gas sensing, microcontroller processing, display, alarm, wireless communication, data logging, power, and reliability concepts into a physical prototype. The supporting technical notes further document the component-level concepts behind the Arduino platform, LCD interface, MQ-series gas sensors, Wi-Fi communication, power supply, and development environment.

The resulting project provides a useful demonstration of practical hardware integration, embedded development, sensor interfacing, system testing, troubleshooting, and engineering project leadership.

25. Author

Vattam Madhu Kumar

Electronics & Communication Engineering

GitHub: <https://github.com/MadhuKumar-003>

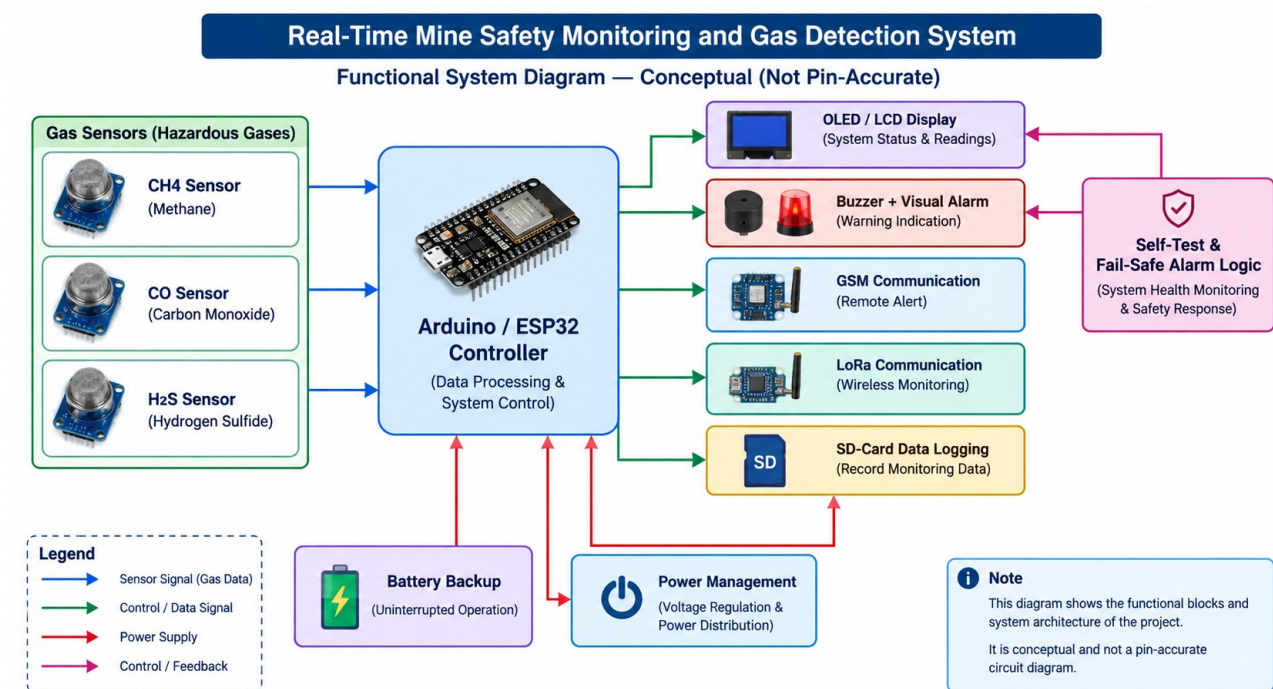
LinkedIn: <https://linkedin.com/in/madhu-kumar-vattam/>

26. Project Image



Actual major diploma project hardware.

27. Functional System Diagram



Conceptual functional diagram - not pin-accurate.